

Tableau Pulse Workshop (2026)

Business Applications Co., Ltd.



Tableau

The World's #1 Analytics Platform

Evolution of Analytics



Pre 2003

Traditional Reporting

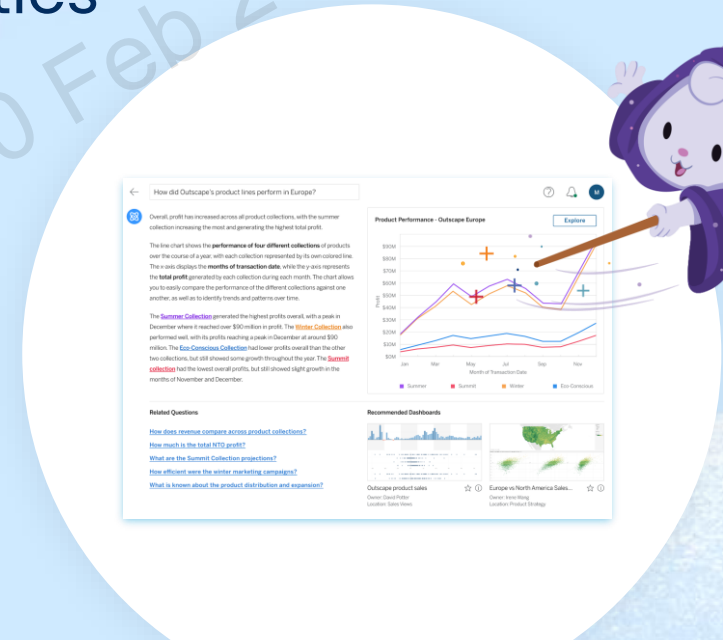
See Data



2004 - 2022

Self-Serve Analytics

See and Understand Data



2023 & Beyond

AI + Analytics for Everyone

See, Understand, and Make
Decisions with Data

Tableau AI extends your Data Experiences

AI infused in your data journey to accelerate your impact

Today's Pulse

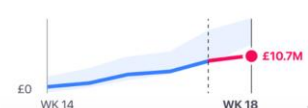
Last updated yesterday
 Last week, the **Average Call Handle Time** spiked to 8.3, dropped by 20.0% compared to last month. On a positive note, it increased by 29.8% compared to the same time last month favorably, 3 unfavorably.

Was this helpful?

Natural Language Insights
 Tableau Pulse

Branch Revenue

£10.7M
 +£0.7M (+7% quarter to date)



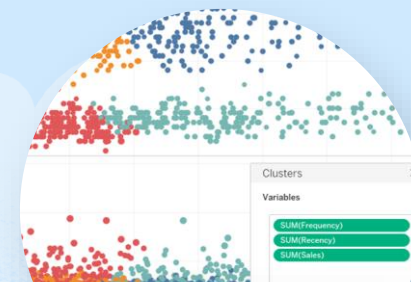
Business Signal Detection
 Tableau Pulse

AI-Assisted Calculations
 Einstein Copilot

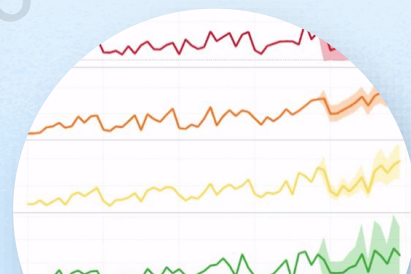
You can use the REGEXP_EXTRACT function to extract part of a string that matches a specific pattern. To extract the postal code from the Customer Mailing Address field, try this:

```
REGEXP_EXTRACT([Customer Mailing Address], "(\\d{5})\\D*$")
```

Clustering
 Tableau Authoring



Forecasting
 Tableau Authoring

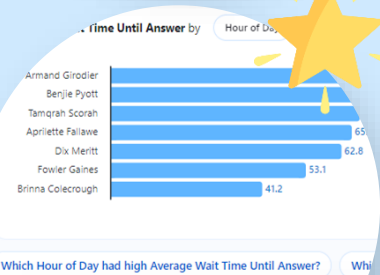


Current Analytics Connection

```
localhost:9004
1 # Import libraries
2 import pandas as pd
3 from collections import Counter
4 import string
5 import re
```

External Computes
 Tableau Authoring

Root Cause Analysis
 Tableau Pulse



Einstein Predictions - Churn

6.71
 Top Predictors
 Negative Interactions +12.25
 Positive Feedback on Social Media -10.52
 Negative Interactions and Invite Event is false +10.45

Predictive AI
 Einstein Prediction



Smart Recommendations
 Tableau Cloud

AI-Assisted Creation
 Tableau Einstein Copilot

show me the most popular flats by reviews
 I will show a bar chart with the flats on the x-axis and the total number of reviews on the y-axis. The flats will be sorted by the

Tableau makes trusted insights accessible to all with AI

Tableau
Pulse

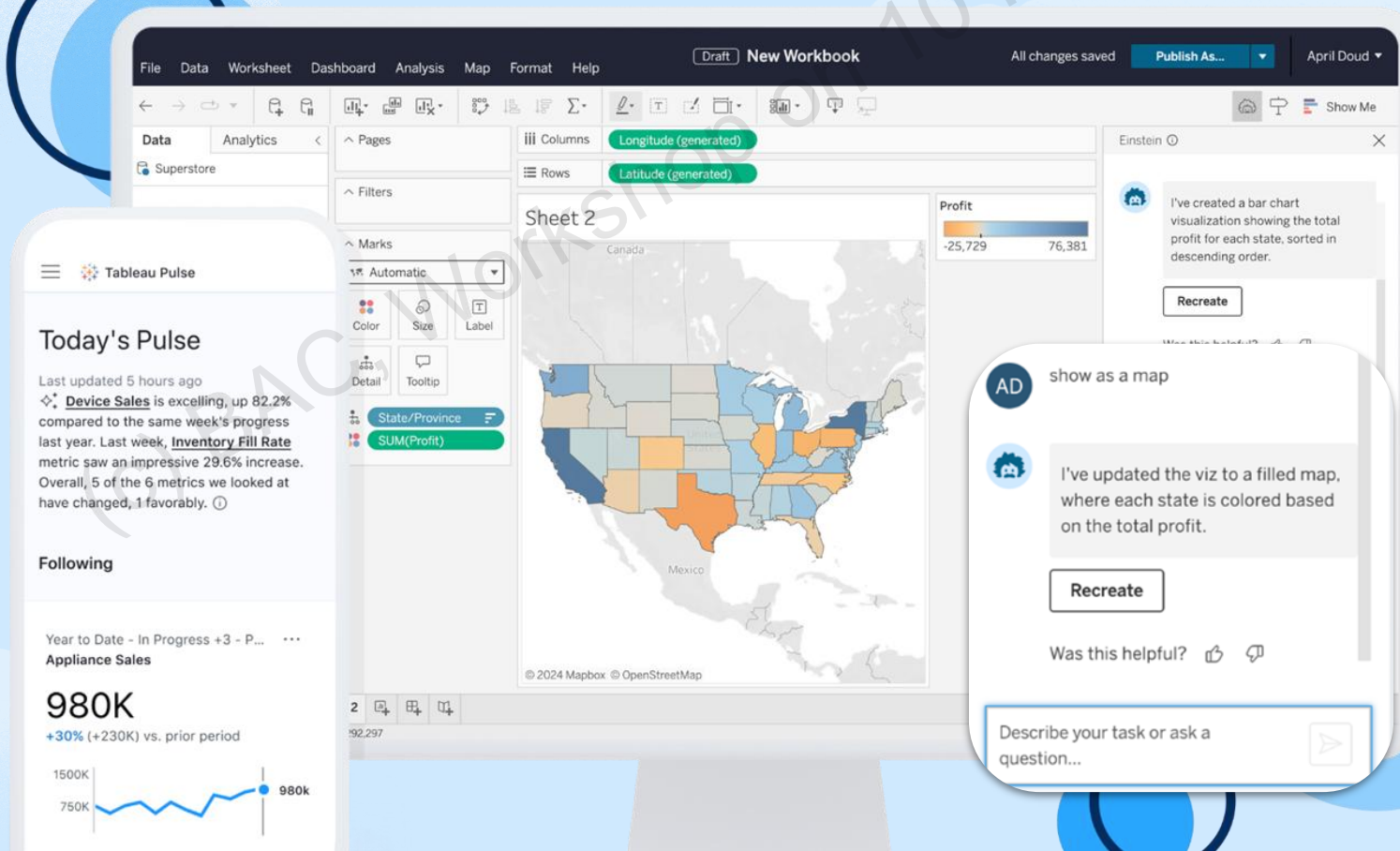


Tableau
Agent

**Which Dashboard is
the most important
dashboard in
a company ?**



Why KPI Dashboard is the most important ?

- Facilitate effective decision-making. KPI dashboards give a consolidated view of key performance metrics across the organization in one screen
- Promote accountability and transparency by clearly displaying progress towards goals and highlighting areas needing attention.

Note: KPIs should be regularly reviewed and adjusted to remain relevant as conditions change

What Are Your KPIs Really Measuring?, Harvard business review
Key Performance Indicator (KPI) Definitions, Gartner

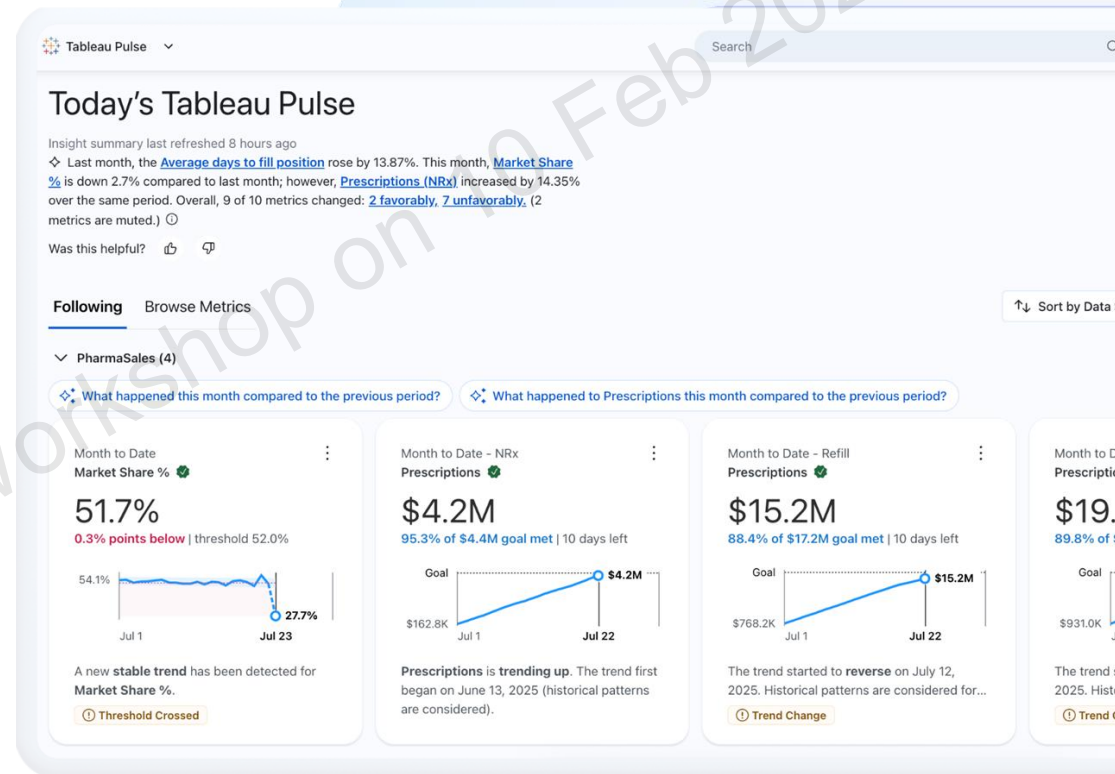
Tableau Pulse

Scale data-driven decisions with AI-powered insights

Stay on top of your business with personalized metrics and proactive insights in your flow of work

Dig deeper into trends and insights with guided exploration and natural language Q&A

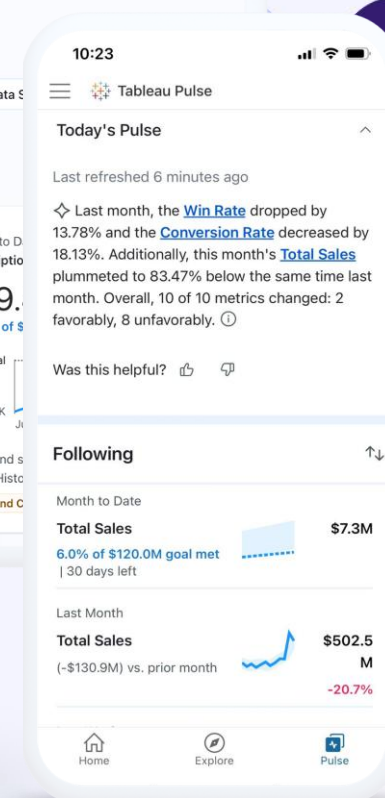
Gain trust in your data with standardized, reusable metrics infused with your business context



JUNIPER
NETWORKS

2K

hours of productivity gained each month



Pulse is designed for how business users want to work with data



1. Alert

Get updates on metrics in your flow of work



2. Understand

Find trends and discover the “why”



3. Explore

Dive deeper and find new insights



4. Follow-up

Ask questions in a conversational interface



5. Share

Collaborate and make data-driven decisions



Alert

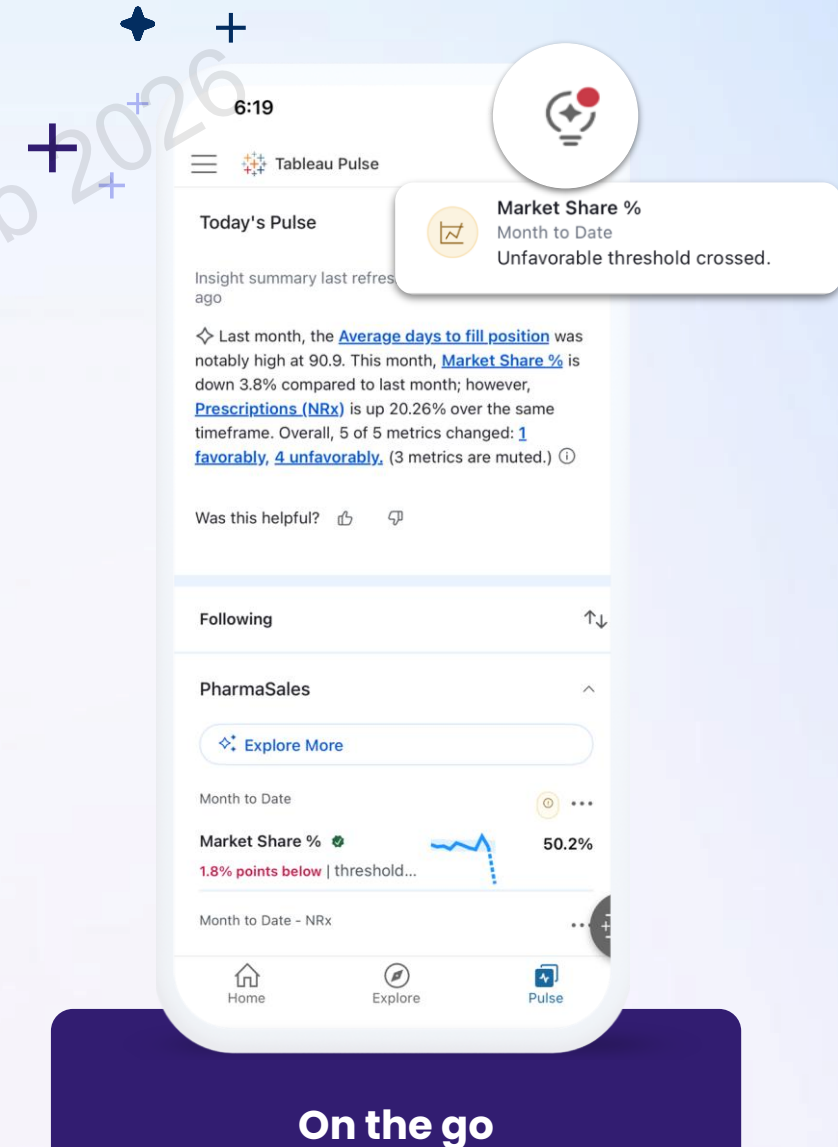
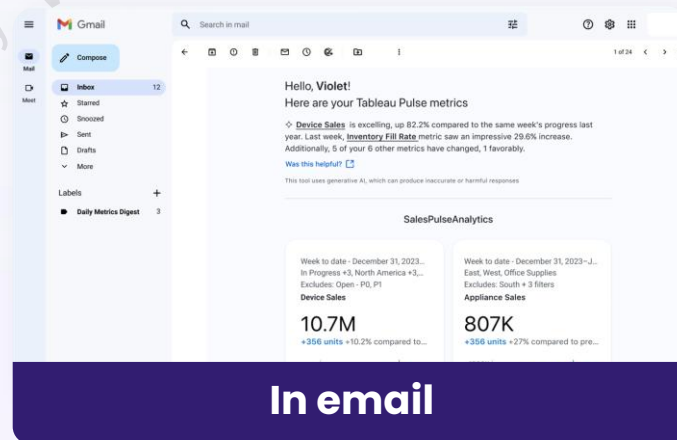
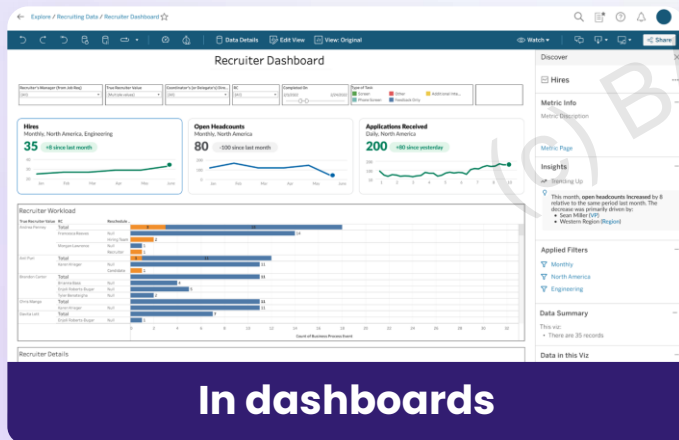
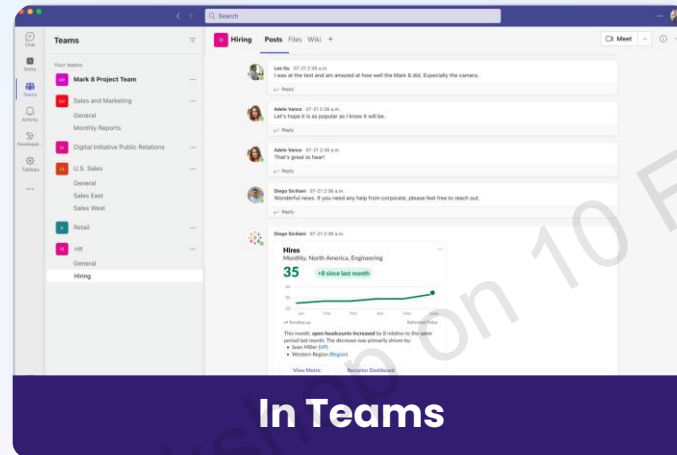
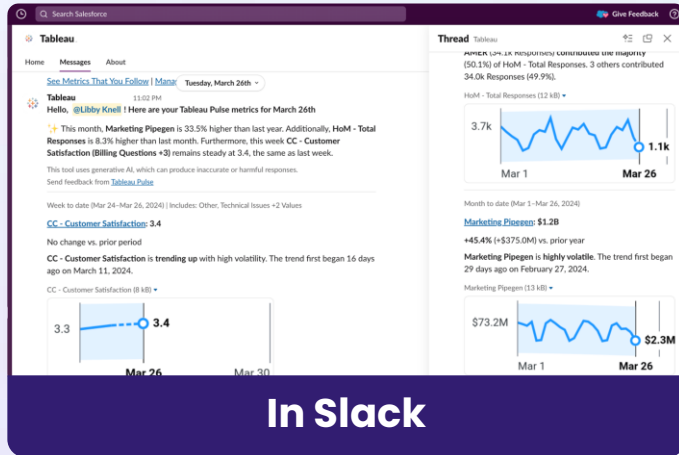
Understand

Explore

Follow-up

Share

Get updates on metrics in your flow of work



Find trends and discover the “why”

Find trends in the data with the help of AI

Today's Tableau Pulse

Insight summary last refreshed 20 hours ago

✧ The monthly [Transaction Comparable \(ratio\)](#) was notably high at 64.8 yesterday; however, the monthly [Average Order \\$](#) was low at \$14.44. Meanwhile, the monthly [Total Revenue](#) saw a modest increase, reaching \$7.4M yesterday. Overall, 6 of 6 metrics changed: [5 favorably](#), [1 unfavorably](#). (3 metrics are muted.) ⓘ

Was this helpful? 👍 🗑

Following Browse Metrics

Chain Restaurant Management (4)

What happened this month compared to the previous period?

Month to Date
Average Wait Time (minutes)

5.1

0.43 minutes (8%) below | threshold 5.5 minutes



Compared to last month, **Average Wait Time (minutes)** increased by 0.39 minutes. Out of the 21 members of State, 18 increased. **Georgia, Utah, and Texas** increase...

Month to Date
Customer Satisfaction

3.0

0.25 (8%) below | threshold 3.2



Compared to last month, **Customer Satisfaction** decreased by 0.28. Out of the 34 members of Menu...

ⓘ Threshold Crossed

Month to Date
Total Revenue

\$5.4M

+64.9% (+\$2.1M) vs. prior period



Bottom contributors by Menu Category are: **Chicken** with \$54.8K (1.0%), **Shake and Desserts** with \$299.3K...

ⓘ Trend Change

See when and where things changed

Top insight about this change | Does Total Revenue have a new trend?

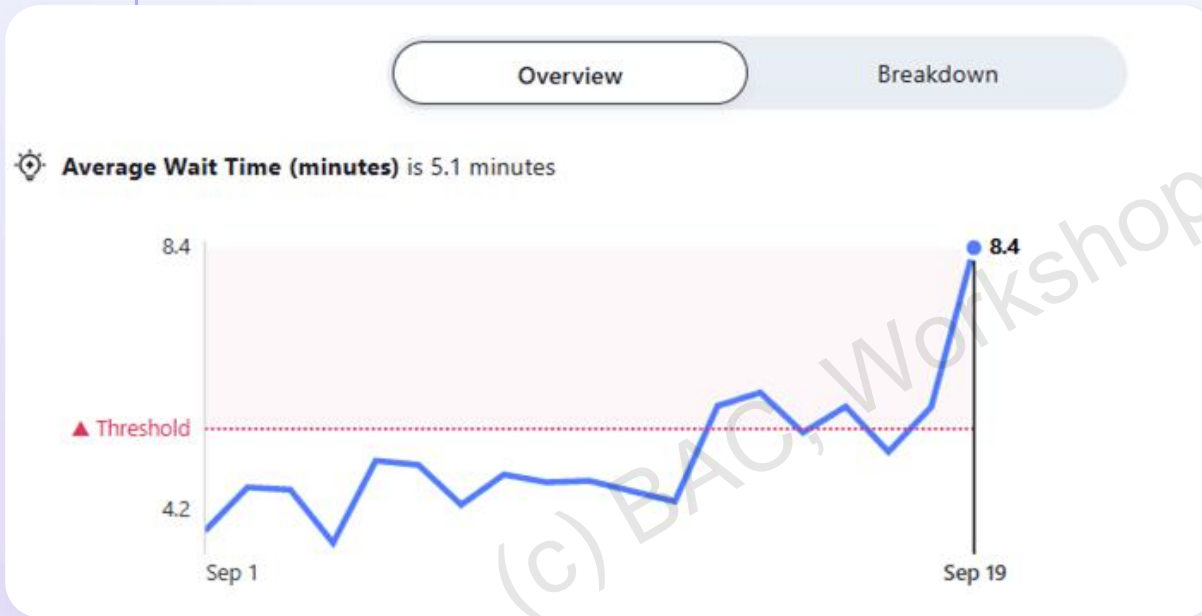
⚙ The trend started to **steepen** on September 13, 2025. Historical patterns are considered for trends.



Was this helpful? 👍 🗑

Dive deeper and find new insights

Get an AI-powered overview of each metric

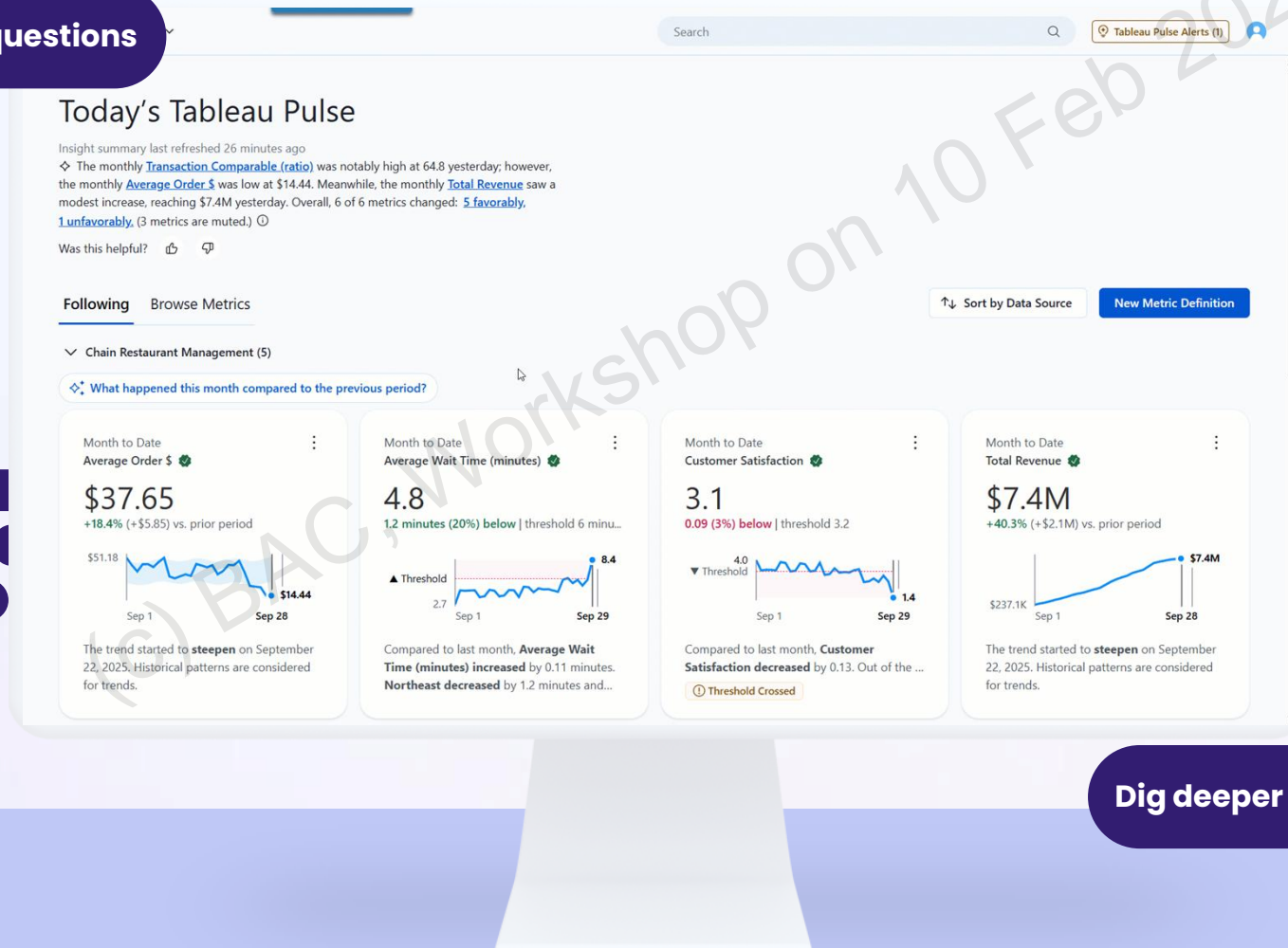


Dig deeper with on-demand filtering



Ask questions in a conversational interface

Start with suggested questions



Dig deeper with conversational exploration*

Alert

Understand

Explore

Follow-up

Share

Collaborate and make data-driven decisions

Add team members to metrics to share insights

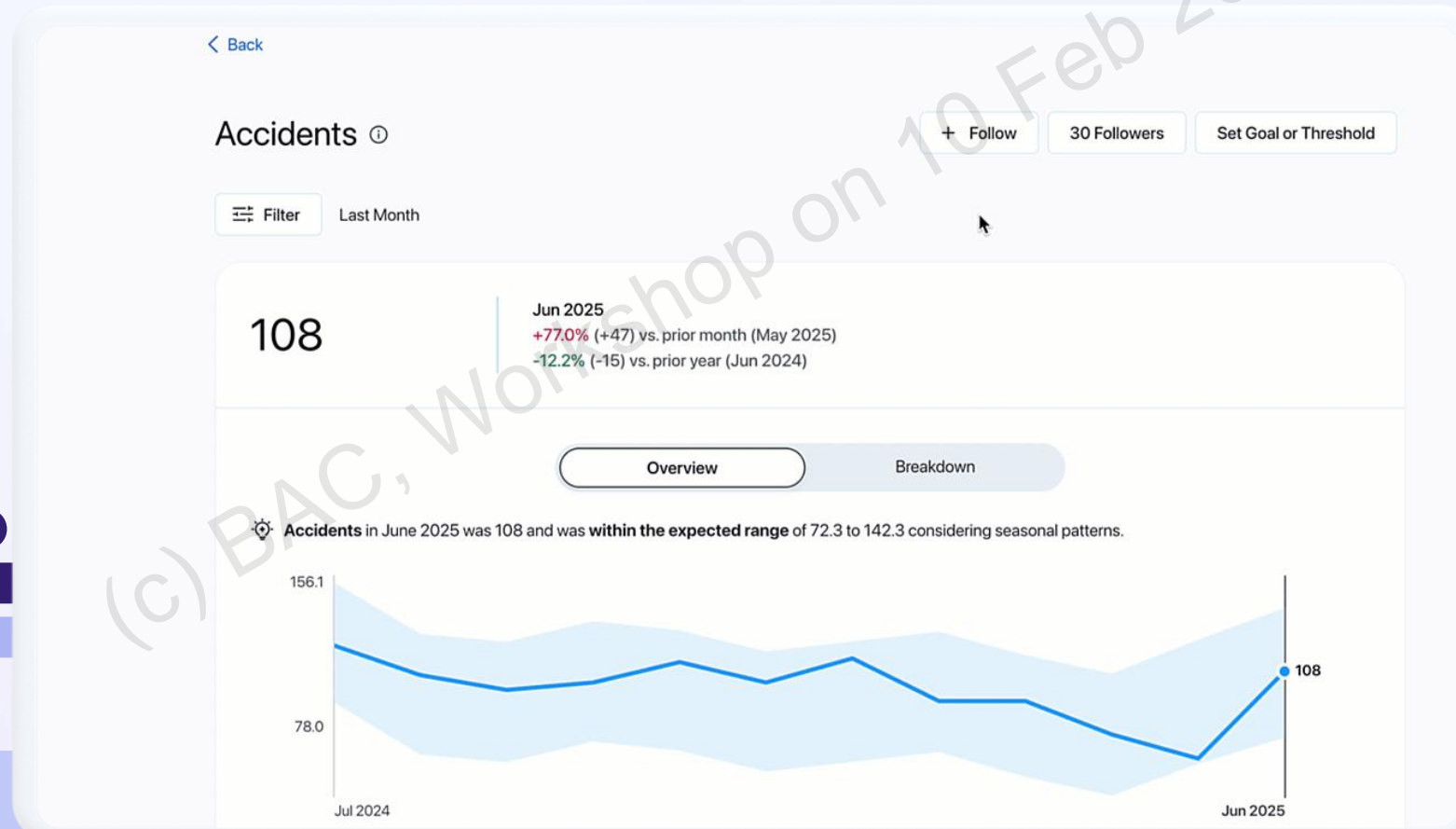


Tableau Pulse delivers a reimagined data experience

Intelligent

Discover

Question | Call Center: What happened this month compared to the previous period? 4:04 PM

✦ This month, the call center saw a 6.4% increase in **Avg Incoming Calls per Agent per Day** [1] and a 6.4% increase in **Incoming Calls** [2]. However, **Total Abandoned Calls** surged by 404.2% [3]. Additionally, **AHT Average Handle Time** increased by 6.5% [4] and **Avg Wait Time (s)** rose by 23.7% [5]. On a positive note, **CSAT - Customer Satisfaction** improved by 4.1% [6].

> Metric Sources (6)

Was this helpful?

✦ What caused the increase in Total Abandoned Calls?

✦ How can we further improve CSAT - Customer Satisfaction?

Question | Who can help me increase call volume? 4:04 PM

✦ To increase call volume, you might want to look at agents like **Valentijn Lambdin**, **Gabriella Galliver**, and **Anabel Grebert** who had the highest increases in **Avg**

Ask a question

This tool uses generative AI, which can produce inaccurate or harmful responses. [Learn more](#)

Personalized

Tableau Pulse

Search

Today's Tableau Pulse

Insight summary last refreshed 13 seconds ago

✦ The monthly **Total Abandoned Calls** reached a high of 87 yesterday. This month's **Avg Wait Time (s)** is 23.69% higher than the same time last month, while **Incoming Calls** is up 6.39% over last month. Overall, 6 of 6 metrics changed: [1 favorably](#), [3 unfavorably](#), [2 neutral or unspecified](#). ⓘ

Was this helpful?

Following Browse Metrics Sort by Data Source

Call Center (6)

What happened this month compared to the previous period?

Month to Date
AHT Average Handle Time
319.9
+6.5% (+19.5) vs. prior period



Month to Date
Avg Incoming Calls per Agent per Day
68.6
+6.4% (+4.1) vs. prior period



Month to Date
Avg Wait Time (s)
103.8
+23.7% (+19.9) vs. prior period



Top contributors by Agent are: **Briana Colecrough** (396.9 s), **Gabriella Galliver** (393.5 s), and **Reta Winscum** (342.9 s).

Compared to last month, **Avg Incoming Calls per Agent per Day** increased by 4.1 calls per agent per day. **Reinwald Lewis**...

Compared to last month, **Avg Wait Time (s)** increased by 19.9 s. Out of the 6 members of **Call Reason**, 6 increased....

Contextual

Tableau Pulse

Today's Pulse

Insight summary last refreshed half a minute ago

✦ The monthly **Total Abandoned Calls** reached a high of 80 yesterday. This month, the **Incoming Calls** is up 1.13% compared to the same time last month, while the **Avg Wait Time (s)** is 26.14% higher than last month. Overall, 6 of 6 metrics changed: [1 favorably](#), [3 unfavorably](#), [2 neutral or unspecified](#). ⓘ

Was this helpful?

Following Sort by Data Source

Call Center

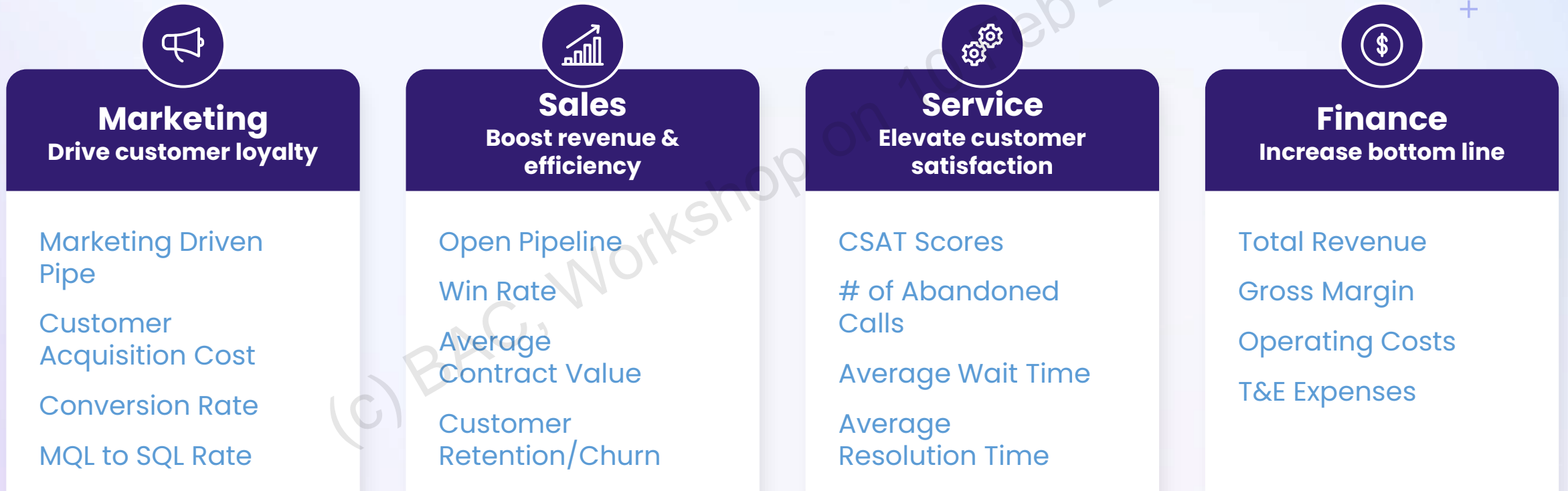
[Explore More](#)

Month to Date

AHT Average Handle Time **319.9**
(+19.5) vs. prior period **+6.5%**

Home Explore Pulse

Tableau Pulse unlocks insights for every department

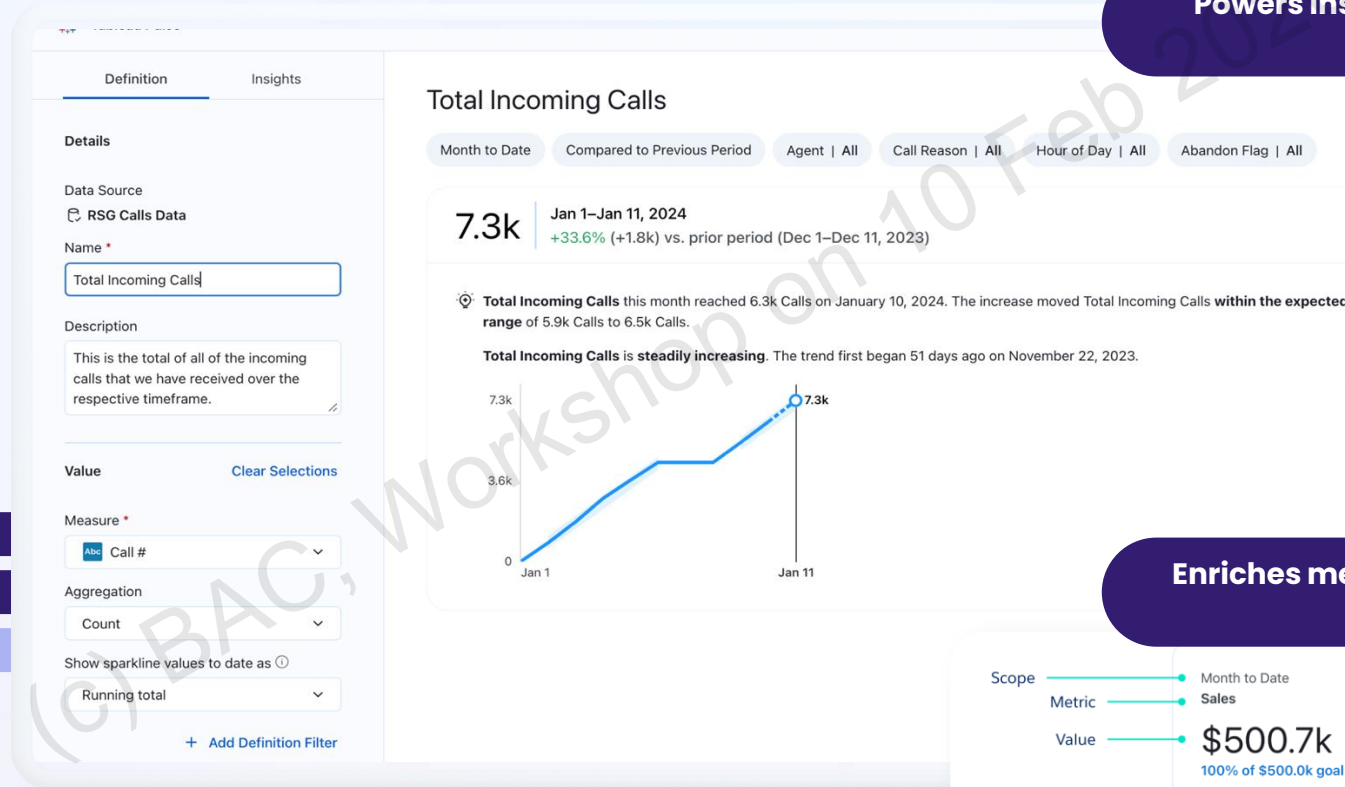


Actionable insights and personalised metrics in your flow of work – when and where you need them

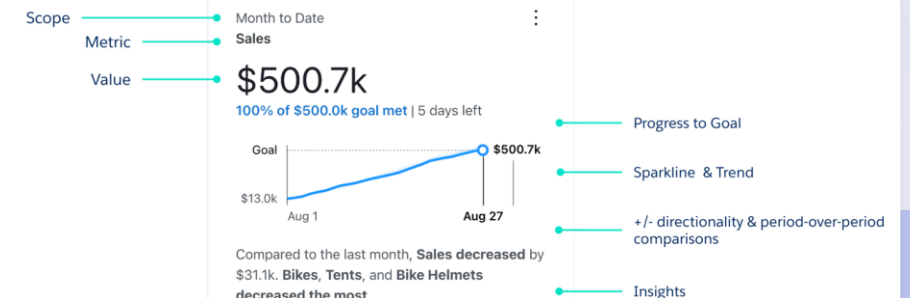
Tableau Pulse: Metrics Layer

Captures the details your business needs

Powers insights with standard metrics definitions



Enriches metrics with real world business context



Enables reuse of metrics across use cases

Components of the Pulse Metrics Layer



Metric Definition

A scalable & manageable source of truth

Measure & aggregation define the metric's data.

Time dimension & temporal level of detail to track & filter the metric over time.

Definitional filters required for the metric to be right, e.g. closed_won = True for "Bookings."

Related dimensions along which the metric can be meaningfully broken down.

Additional metadata & relationships like "up is good", or pointers to other metrics or analytics content to help contextualize the metric.



Metric

An easy, understandable analytics artifact

Metric value provides users "the number" to orient on.

Sparkline & trend show how the metric is performing over time.

Scope across time period and cut by dimension.

Insights create scannable summaries for users.

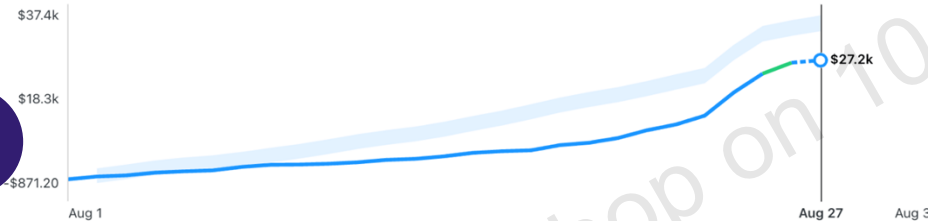
Actions can be taken against that metric, integrated with the tools available across the portfolio.

Tableau Pulse: Insights Platform

Contextual, personalized, and intelligent insights across apps

Creates automated,
proactive questions

⚙️ **Returns** this month reached \$26.6k on August 26, 2024. The unexpected small increase moved Returns **below the expected range** of \$32.8k to \$36.3k, considering seasonal patterns.
The trend for **Returns** changed 4 days ago and is now **increasing at a slower rate**.



Shows the where, when, and why

Discover Top Insights



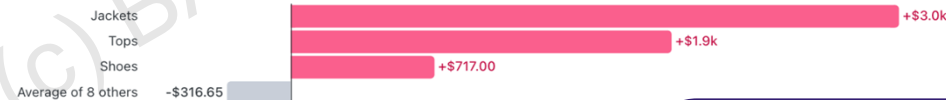
Which Subcategory had very high Returns?

Does Returns have a new trend?

Which Product Type decreased the most?

Top insight about this change | Which Subcategory increased the most?

⚙️ Compared to the last month, **Returns increased** by \$3.1k. **Jackets, Tops, and Shoes increased the most**.



Automatically identifies drivers, trends,
and outliers

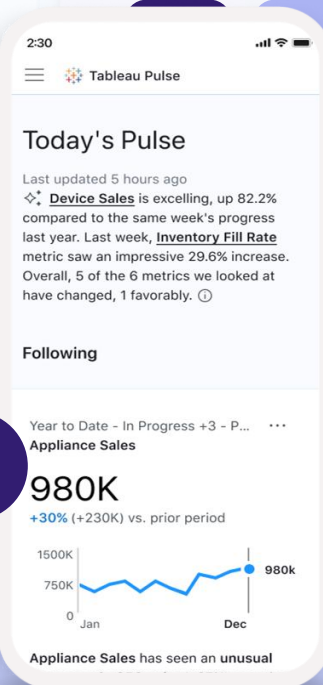


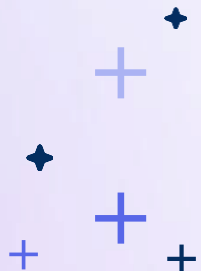
Tableau Pulse:

Capabilities by Edition & License Type

Editions	Capability	Viewer	Explorer	Creator
Standard & Enterprise Editions	Metrics Layer - Create metric definition		✓	✓
	Metric creation from Data Guide		✓	✓
	Lineage via Tableau Catalog		✓	✓
	Insights platform AI	✓	✓	✓
	Insights summaries AI	✓	✓	✓
	Insights & metrics in email, Slack, Teams, & mobile app	✓	✓	✓
	Custom calendar	✓	✓	✓
	Goals & Thresholds	✓	✓	✓
	Alerts	✓	✓	✓
	Global currencies & languages	✓	✓	✓
	Data-based actions on charts	✓	✓	✓
Tableau+ only	Enhanced Q&A AI	✓	✓	✓
	Correlated insights AI	✓	✓	✓
	Time-based forecasting AI	✓	✓	✓

Note: This is not an intended to be an exhaustive list of all Pulse capabilities and features. However, it does list all Tableau+ exclusive features.

Tableau Pulse Requirements



(c) BAC, Workshop on 10 Feb 2026

Data source requirements

Published data source

- Live, Extract, Bridge, and Virtual Connections all supported

Metric definition prerequisites

- A measure
- A time dimension
- At least one dimension column
- Standard English Column Name

Data source requirements

- A single published data source. You can't connect to a data source that is embedded in a workbook, and you can't connect to multiple data sources or use data blending, unless you combine the data before publishing the data source.
- You have the Connect and View permission capabilities for the data source.

Data source features requirements

- A measure to be aggregated as a sum, average, median, maximum, or minimum or a dimension to be aggregated as a count or count (distinct).
- A time dimension for the metric's time series. Tableau Pulse monitors data over time, so single point-in-time values won't produce a valid metric.
- At least one dimension that can be used to filter the data and insights.

Naming Convention

- Use title-style capitalization rather than all caps.
- Use spaces rather than hyphens or underscores to separate words.
- Don't use special characters.
- Don't use Boolean fields that appear as true or false.
- Don't use fields that consist of IDs.

Requirements for data source goals

- Goals can be static or user-defined
- Include the field that specifies the goal value as part of the data source that the definition connects to
- If you want a finer-grained goal to display when users apply dimensional filters on metrics, include that level of granularity in your goal data

Recommended Dataset

A dimension (for drill-down or breakdown), should have low cardinality

Date Prefer Daily	Low-Cardinality	Metric-Based				
Date	Region	Product	Measure 1	Measure 2	Measure 3	Measure 4
2024-01-01	North	Product A	1,000	2,500	3,150	4,200
2024-01-01	Central	Product A	2,000	2,600	3,750	4,000
2024-01-01	South	Product A	3,000	2,700	4,250	3,800
2024-01-01	North	Product B	1,500	2,550	3,250	3,500
2024-01-01	Central	Product B	2,500	2,650	3,950	3,600
2024-01-01	South	Product B	3,500	2,750	4,450	3,950

Metric Definitions Conclusion

Basic definition prerequisites

- A measure
- A time dimension
- At least one filter (with low-cardinality recommended)
- Standard English column name

Optional definition

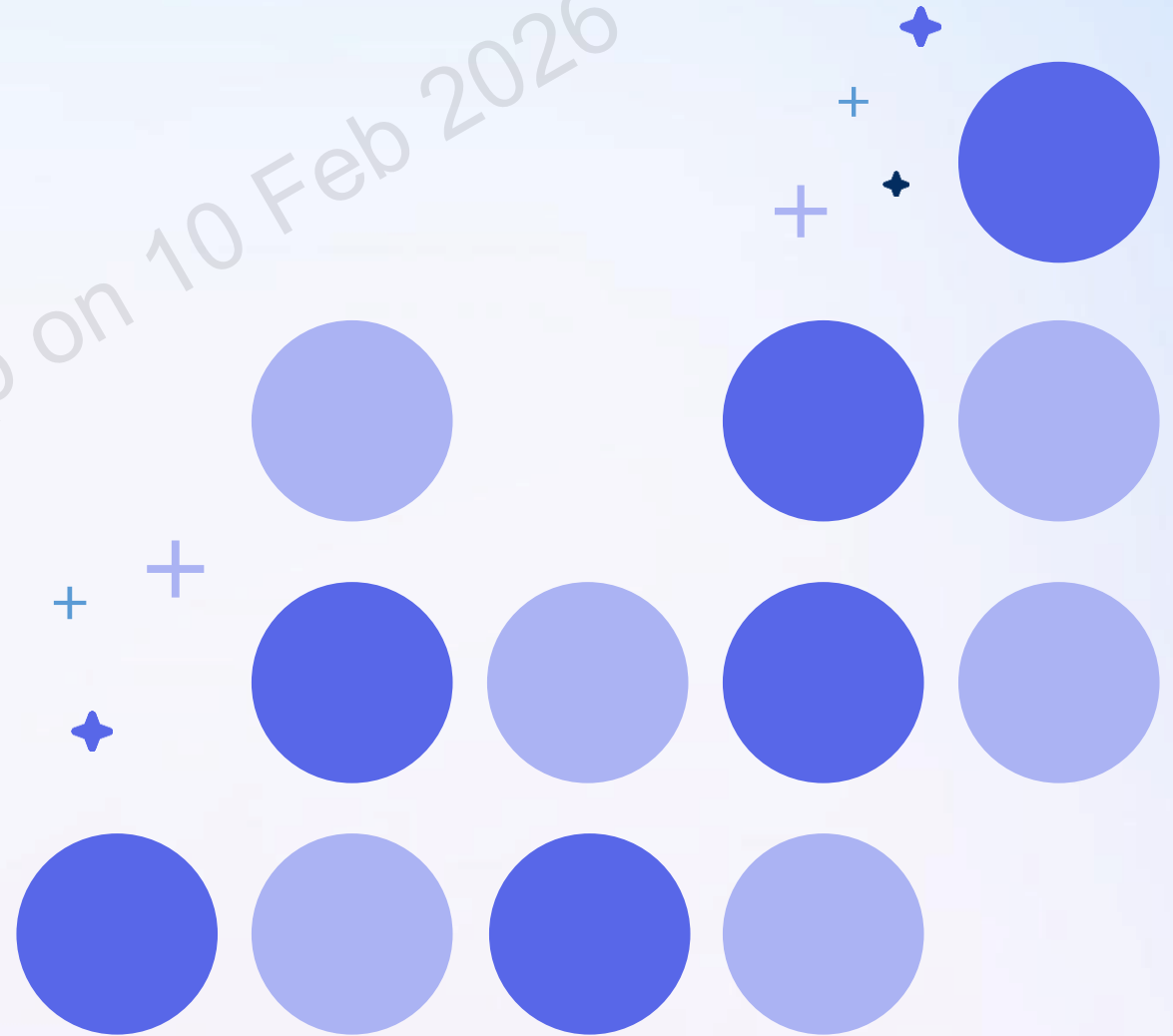
- Definition filter
- Measure aggregation (Avg, Sum, Min, Max) / sparkline display adjustments
- Number format and unit expression
- Value up/down good/bad
- Goal

WORKSHOP

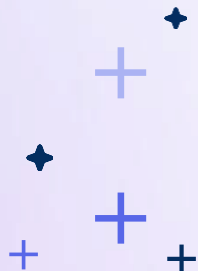
Introduction to Tableau Pulse



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How to unlock maximum value from Tableau Pulse ?



Recommended Effective Data Sets

Data changes frequently

- Daily is ideal
- Daily insights need daily data
- Monthly or longer is less successful

Data must be lean and agile

- Large data sets lose relevance
- Data contains only key dimensions

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Baseline Metric

Baseline

Revenue

Time-Variant

- Week to Date
- Month to Date
- Quarter to Date
- Year to Date

Automated By Tableau Pulse

- VS Yesterday
- VS Last Week
- VS Last Month
- VS Last Quarter
- VS Last Year

Filtering Columns

Automated By Tableau Pulse

- By Product Group
- By Region/Area

Composite Metric Super-Charging AI Tools

Composite metrics provide a more complete picture than single metrics.

They combine different factors, leading to better decisions, capture business complexity, reduce bias, and clearer communication of complex information.

Analysis Skill
Needed



Time-Variant

- Week to Date
- Month to Date
- Quarter to Date
- Year to Date
- VS Yesterday
- VS Last Week
- VS Last Month
- VS Last Quarter
- VS Last Year

Automated
By Tableau Pulse

Baseline

Revenue, Quantity, Order

Derived
Metric

Rev. per Quantity = Revenue/Quantity
Rev. per Order = Revenue/Order

Filtering
Columns

- By Product Group
- By Region/Area

Automated
By Tableau Pulse

Critical Thinking / Design Thinking makes AI Tools more valuable.

Baseline	Revenue, Quantity, Order	Critical Thinking Needed		
Derived Metric	Rev. per Quantity = Revenue/Quantity Rev. per Order = Revenue/Order		Time-Variant	Automated By Tableau Pulse
Polarity Balancing	Revenue – Cost = Profit % Profit = Profit / Revenue		• Week to Date • Month to Date • Quarter to Date • Year to Date	• VS Yesterday • VS Last Week • VS Last Month • VS Last Quarter • VS Last Year
Efficiency	Using Amount of Resource / Number of Staffs Revenue / Staffs, Revenue / Resource Utilization			
Customer	Revenue / Customer Revenue / Customer Lifetime Value Revenue / Frequent Customer One-Time Customer		Filtering Columns	Automated By Tableau AI
Comparison	VS. Target / VS. Previous Year Revenue vs Target / Revenue vs Last Year + 10%			• By Product Group • By Region/Area
Peer Comparison	Ranking of Revenue within Peer (Dimension), Ratio of Revenue within (Dimension)			

KPI Card Design

Basket Size (Amount)

Description : *The average amount a customer buys per transaction.*

Calculation :

Total Sales Amount / Number of Transactions

Supporting KPI:

Conversion Rate, Traffic, Average Discount %, Return Rate, Gross Margin %

KPI Ownership:

Mr. AAA BBB

KPI Card Design

Basket Size (Amount)

Increasing Meaning

Customers are buying **more value per visit**

Possible drivers: better bundling, upsell/cross-sell, premium mix, pricing increase, fewer discounts

Decreasing Meaning

Customers are buying **less per visit**

Possible drivers: weaker merchandising, stock-outs in attach items, heavier discounting on low-priced SKUs.

Action when High

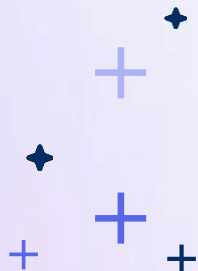
- **Validate quality:** check **Gross Margin %** and **Return Rate** (avoid “high sales, low profit”)
- **Scale what works:** replicate bundles, recommendations, in-store scripts, PDP modules, checkout add-ons
- **Protect supply:** ensure availability of attach items and hero SKUs

Action when Low

- **Find the break point:** category, store, channel, customer segment, campaign
- **Fix attach rate:** introduce bundles, “buy more save more”, add-on placement, checkout cross-sell
- **Check inventory:** stock-outs often kill basket growth (especially complementary SKUs)
- **Check price/discount strategy:** are you discounting the wrong products and shrinking spend?

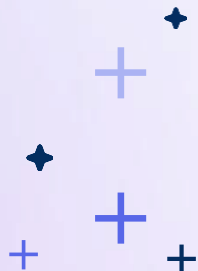
WORKSHOP

Critical Thinking



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Don't misunderstand Tableau Pulse



Using Tableau's Super Store Dataset Which are best metrics for them ?

Base Metrics	Compound Metrics	Time Metrics	Customer Metrics	Product Metrics	Operations Metrics
Sales = SUM([Sales])	Profit Ratio (Margin %) = SUM([Profit]) / SUM([Sales])	Monthly Sales = SUM([Sales]) (by OrderYearMonth)	Revenue per Customer = SUM([Sales]) / COUNTD([CustomerID])	Sales Mix % (Category) = SUM([Sales]) / TOTAL(SUM([Sales]))	Ship Days = DATEDIFF('day',[OrderDate],[ShipDate])
Profit = SUM([Profit])	ASP (Unit Price) = SUM([Sales]) / SUM([Quantity])	MoM Sales Growth % = (SUM([Sales]) - LOOKUP(SUM([Sales]),-1)) / LOOKUP(SUM([Sales]),-1)	Profit per Customer = SUM([Profit]) / COUNTD([CustomerID])	Profit Mix % = SUM([Profit]) / TOTAL(SUM([Profit]))	Avg Ship Days = AVG([Ship Days])
Quantity = SUM([Quantity])	Profit per Unit = SUM([Profit]) / SUM([Quantity])	YoY Sales Growth % = (SUM([Sales]) - LOOKUP(SUM([Sales]),-12)) / LOOKUP(SUM([Sales]),-12)	Orders per Customer (Frequency) = COUNTD([OrderID]) / COUNTD([CustomerID])	Unprofitable Sales % = SUM(IF [Profit]<0 THEN [Sales] END) / SUM([Sales])	Orders by ShipMode = COUNTD([OrderID]) (by ShipMode)
Orders = COUNTD([OrderID])	Sales per Order (AOV) = SUM([Sales]) / COUNTD([OrderID])	YTD Sales = RUNNING_SUM(SUM([Sales])) (restart yearly)	Customer Lifetime Sales (LOD) = { FIXED [CustomerID] : SUM([Sales]) }	Loss Orders = COUNTD(IF [Profit]<0 THEN [OrderID] END)	Profit per ShipMode = SUM([Profit]) / COUNTD([OrderID])
Order Lines = COUNT([RowID])	Profit per Order = SUM([Profit]) / COUNTD([OrderID])	YTD Profit = RUNNING_SUM(SUM([Profit])) (restart yearly)	Customer Lifetime Profit (LOD) = { FIXED [CustomerID] : SUM([Profit]) }	Product Profitability (by Product) = SUM([Profit])	ShipMode Mix % = COUNTD([OrderID]) / TOTAL(COUNTD([OrderID]))
Discounts (Avg) = AVG([Discount])	Items per Order = SUM([Quantity]) / COUNTD([OrderID])	Rolling 3M Sales = WINDOW_AVG(SUM([Sales]),- 2,0)	Customer Margin % (LOD) = { FIXED [CustomerID] : SUM([Profit]) } / { FIXED [CustomerID] : SUM([Sales]) }	Top-N / Rank Products (Sales) = RANK(SUM([Sales]))	Avg Discount by ShipMode = AVG([Discount]) (by ShipMode)
Distinct Categories = COUNTD([Category])	Weighted Discount Rate = SUM([Sales]*[Discount]) / SUM([Sales])	Rolling 6M Sales = WINDOW_AVG(SUM([Sales]),- 5,0)	New Customer Flag = DATETRUNC('month',[OrderDate]) = DATETRUNC('month',{ FIXED [CustomerID] : MIN([OrderDate]) })	Margin by Discount Band = SUM([Profit]) / SUM([Sales]) (slice by band)	Orders with Slow Shipping % = SUM(IF [Ship Days]>X THEN 1 END)/COUNTD([OrderID])
Distinct SubCategories = COUNTD([SubCategory])	Profit Contribution % = SUM([Profit]) / TOTAL(SUM([Profit]))	Sales Index (Base=100) = SUM([Sales]) / LOOKUP(SUM([Sales]),- (INDEX()-1)) * 100 (concept)	Returning Customer Flag = NOT([New Customer Flag])	Pareto Cumulative Sales % = RUNNING_SUM(SUM([Sales])) / TOTAL(SUM([Sales])) (sorted)	Avg Order Value (true order grain) = AVG({ FIXED [OrderID] : SUM([Sales]) })
Distinct Regions = COUNTD([Region])	Sales Contribution % = SUM([Sales]) / TOTAL(SUM([Sales]))	Avg Monthly Sales = WINDOW_AVG(SUM([Sales]))	Customer AOV (LOD) = { FIXED [CustomerID] : SUM([Sales]) } / { FIXED [CustomerID] : COUNTD([OrderID]) }	SubCategory Margin % = SUM([Profit]) / SUM([Sales]) (by SubCategory)	Avg Margin per Order (order grain) = AVG({ FIXED [OrderID] : SUM([Profit]) } / { FIXED [OrderID] : SUM([Sales]) })
Distinct ShipModes = COUNTD([ShipMode])	Profit per Customer = SUM([Profit]) / COUNTD([CustomerID])		Customer Profitability Segment (by LOD) = { FIXED [CustomerID] : SUM([Profit]) }		

You are Tableau Pulse Expert You can do everything in Tableau Pulse

How Tableau Pulse can help _____

How Tableau Pulse Help People Lose Weight ?

(c) BAC, Workshop on 10 Feb 2026

From Bottom Up
How can Tableau Pulse Help... ?

From Top Down
How can Tableau Pulse Help... ?

Key Takeaway #1

- Tableau Pulse delivers insights faster and more consistently.
- Strong KPIs and reliable data foundation make those insights meaningful.
- Critical Thinking connects insights to actions

Key Takeaway #2

Outcome set direction. **Process** is execution.
People need motivation,
and **Tool** helps monitor progress.
Everything starts with **data**.

That's how AI creates real impact.